



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

● Experiment No.: 5

1. **Title:** Implement localStorage and sessionStorage to save user preferences like theme settings.
2. **Software/Tools Required:** VS Code, JavaScript,html.
3. **Theory:**
 - 1) Web Storage allows websites to store data in the browser using key-value pairs.
 - 2) localStorage stores data persistently and remains available even after closing the browser.
 - 3) sessionStorage stores data only for the current browser session.
 - 4) setItem() is used to save the selected theme and storage type.
 - 5) getItem() retrieves previously saved theme and storage information.
 - 6) removeItem() removes stored preferences.
 - 7) The program provides Light, Dark, and Blue themes using CSS classes.
 - 8) classList.remove() and classList.add() are used to change the theme dynamically.
 - 9) getElementById() accesses the required HTML elements.
 - 10) innerText displays the current theme, storage type, saved value, and messages.
 - 11) loadPreference() automatically retrieves saved preferences when the page loads.
 - 12) savePreference() stores the selected theme in either localStorage or sessionStorage.
 - 13) removePreference() clears the saved preferences and resets the theme to Light.
 - 14) The program demonstrates web storage and dynamic DOM manipulation.



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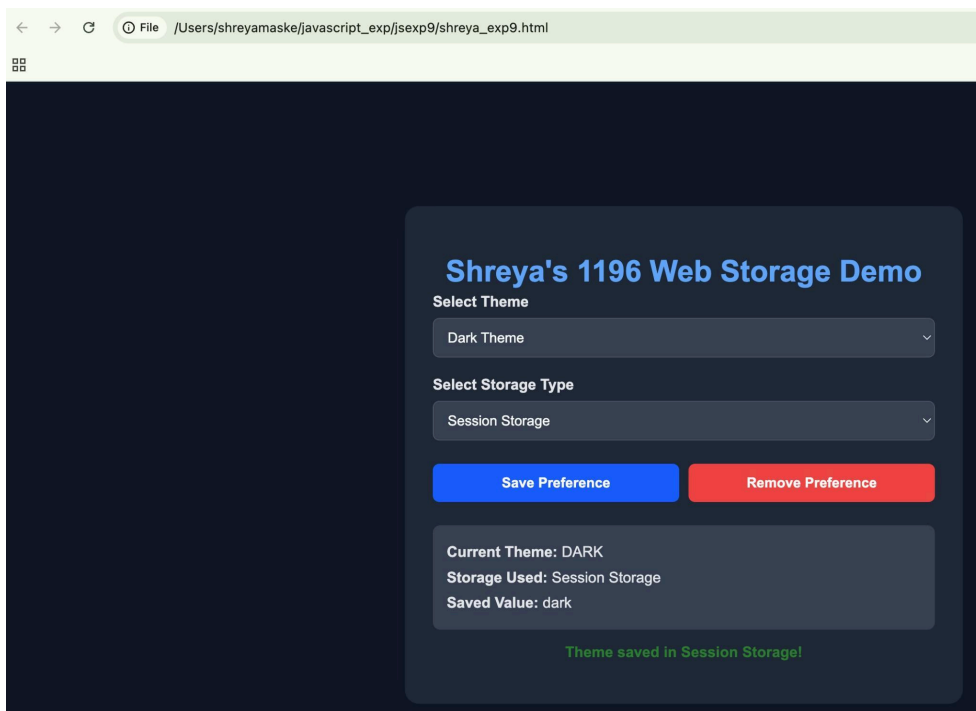
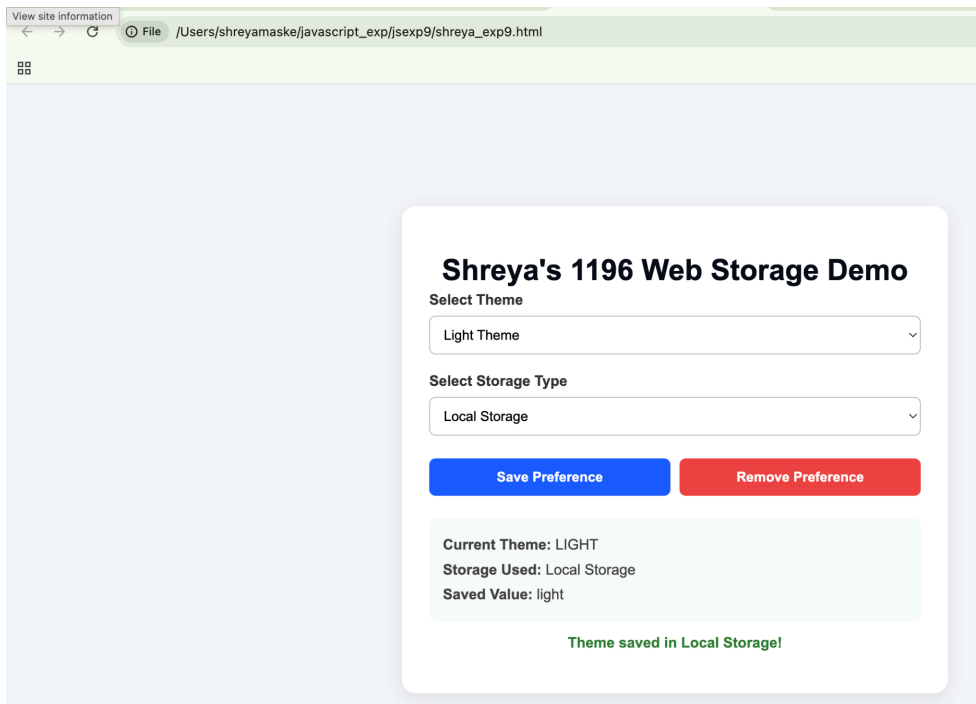
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4. Output :



5. Result/Conclusion:

Thus, local Storage and session Storage were successfully implemented to store and retrieve user theme preferences. The selected theme was applied dynamically, displayed on the webpage, automatically loaded when available, and removed successfully using JavaScript



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Case Study No.: 7

- 1. Case Study Title:** To demonstrate events handling in javascript, a) access and manipulate from elements using javascript. b) validate user input (eg. email, empty, fields) c) use various form events (submit, change, focus).

Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Shreya's (1196) - Event Handling</title>
<style>
body {
font-family: Arial, sans-serif;
background-color: #e8f0fe;
padding: 40px;
}
.form-box {
width: 450px; margin:
auto;
background-color: white; padding: 25px;
border-radius: 10px;
box-shadow: 0 4px 12px gray;
}
h2 {
text-align: center;
}
```



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```
label {
    display: block;
    margin-top: 15px;
    font-weight:
    bold;
}
input, select {
    width: 100%;
    padding: 10px;
    margin-top:
    5px;
    box-sizing: border-box;
}
input:focus,  select:focus
{ border: 2px solid
blue; outline: none;
}
.error {
    color: red;
    font-size: 13px;
}
button {
    width: 100%;
    margin-top:
    20px; padding:
    10px;
    background-color: #2563eb;
    color: white;
    border: none;
```




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```
#message {  
    text-align:  
        center;  
    font-weight: bold;  
    margin-top: 15px;  
}  
  
</style>  
</head>  
<body>  
<div class="form-box">  
    <h2>Shreya's student Registration Form</h2>  
    <form id="studentForm">  
        <label>Name:</label>  
        <input type="text" id="name">  
        <div id="nameError" class="error"></div>  
        <label>Email:</label>  
        <input type="text" id="email">  
        <div id="emailError" class="error"></div>  
        <label>Course:</label>  
        <select id="course">  
            <option value="">Select Course</option>  
            <option value="CSE">Computer  
Science</option>  
            <option value="IT">Information  
Technology</option>  
            <option value="ECE">Electronics</option>  
        </select>  
        <div id="courseError" class="error"></div>  
        <button type="submit">Submit</button>  
    </form>
```

<p id "message"></p>



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```
</div>

<script>

    // Access form elements
    let form =
document.getElementById("studentForm");

    let name = document.getElementById("name");
    let email = document.getElementById("email");
    let course = document.getElementById("course");
    // Focus event
    name.addEventListener("focus", function() {
        name.style.backgroundColor = "#eef4ff";
    });
    email.addEventListener("focus", function() {
        email.style.backgroundColor = "#eef4ff";
    });
    // Change event
    course.addEventListener("change", function() {
document.getElementById("courseError").innerText =
        "Selected Course: " + course.value;
    });

    // Submit event
    form.addEventListener("submit", function(event)
{
        event.preventDefault()
        ; let valid = true;
        // Name validation
        if (name.value.trim() === "") {
            document.getElementById("nameError").innerText
```



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```
        "Name cannot be empty";
        valid = false;
    } else {
document.getElementById("nameError").innerText =
"";
    }
    // Email validation
    let emailPattern =
/^^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (email.value.trim() === "") {
document.getElementById("emailError").innerText =
        "Email cannot be empty";
        valid = false;
    }
    else if (!emailPattern.test(email.value)) {
document.getElementById("emailError").innerText =
        "Enter a valid email";
        valid = false;
    }
    else {
document.getElementById("emailError").innerText =
"";
    }
    // Course validation
    if (course.value === "") {
```



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```
document.getElementById("courseError").innerText =
    "Please select a course";
    valid = false;
}
// Final result
if (valid) {
document.getElementById("message").innerText =
    "Registration submitted
successfully!";

document.getElementById("message").style.color =
"green";
}
else {

document.getElementById("message").innerText =
    "Please correct the errors.";
document.getElementById("message").style.color =
"red";
}
});
</script>
</body>
</html>
```



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3. Output:

Shreya's student Registration Form

Name:

Email:

Course:

Shreya's student Registration Form

Name:

Email:

Course:

Selected Course: CSE

Registration submitted successfully!

4. Result/Conclusion:

Thus, JavaScript events were successfully handled using **focus**, **change**, and **submit**, and form elements were accessed, manipulated, and validated using JavaScript.